

Project Source Code Notebook

Author: Anumari Jyothi
Department: Department of Statistics
Institution: Pondicherry University
Platform: IBM SkillsBuild Internship Project

Source Code Implementation Script

The python script below establishes the end-to-end data processing, feature split validation, and predictive machine learning execution framework.

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

print("Initializing Framework for Anumari Jyothi...")
print("Department of Statistics, Pondicherry University\n")

# Load Dataset
try:
    data = pd.read_csv('heart.csv')
    print("Dataset Loaded Successfully. Shape:", data.shape)
except FileNotFoundError:
    print("Creating synthetic trial dataset for demonstration...")
    np.random.seed(42)
    dummy_data = np.random.randint(50, 200, size=(100, 13))
    dummy_target = np.random.randint(0, 2, size=(100, 1))
    data = pd.DataFrame(np.append(dummy_data, dummy_target, axis=1),
                        columns=['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs',
                                'restecg', 'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'])

X = data.drop(columns='target', axis=1)
y = data['target']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, stratify=y, random_state=2)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

X_train_prediction = model.predict(X_train)
training_data_accuracy = accuracy_score(X_train_prediction, y_train)
print(f'Model Training Data Accuracy: {training_data_accuracy * 100:.2f}%')
```

```
X_test_prediction = model.predict(X_test)
test_data_accuracy = accuracy_score(X_test_prediction, y_test)
print(f'Model Testing Data Accuracy: {test_data_accuracy * 100:.2f}%')
```